# Conveyor Motor Control System

PLC program, HMI and simulation documentation

Siemens S7-1200 (CPU 1212C DC/DC/DC) • TIA Portal V21 • S7-PLCSIM V21 • KTP700 Basic PN

| Item | Detail |
|---|---|
| Project | Conveyor_Control_Version 2 |
| PLC | S7-1200, CPU 1212C DC/DC/DC (PLC_1), LAD, function-block architecture |
| HMI | KTP700 Basic PN, two screens, S7 connection to PLC_1 |
| Engineering / simulation | TIA Portal V21, S7-PLCSIM V21, WinCC Basic runtime simulation |
| Network | PLC 192.168.0.1 \| HMI 192.168.0.2 |
| Author | Biraj |

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| 2 | Hardware and wiring | Wiring diagram, I/O assignment |
| 3 | PLC program | Blocks, PLC tags, FB1 interface, DB10, OB1 call, FB1 ladder networks 1-9 |
| 4 | PLC simulation evidence | PLCSIM fault test, online OB1 / FB1, watch table |
| 5 | HMI | Connection, HMI tag table, screen designs, button events, animations, runtime captures |
| 6 | Operating sequences | Start, stop, emergency stop, fault, reset, timer auto-stop |
| 7 | Design notes | Constraints, decisions and points to be aware of |

# 1. System overview

This project implements the control of a single conveyor motor with four monitored fault conditions, a latched main fault, a run timer that stops the motor automatically, and a two-screen operator panel. The PLC program is written in ladder logic inside one function block (**FB_ConveyorMotor**) called from OB1. The HMI runs on a KTP700 Basic PN and communicates with the PLC over an S7 connection. All testing was done virtually with S7-PLCSIM and the WinCC runtime simulation.

## Functional scope

- Motor start / stop from physical push buttons **and** from the HMI (separate command bits).
- Four fault inputs (overcurrent, overvoltage, belt trip, motor temperature), each with its own indicator lamp.
- Latched **Main Fault** (SET/RESET): any fault trips and shuts the motor down; Reset button or HMI RESET clears it.
- Motor run timer (TON, 25 s preset): the motor stops automatically when the timer completes.
- HMI timer display in seconds (millisecond TIME value converted to DInt).
- Emergency stop button on both HMI screens (latching stop command).
- Two HMI screens: *Conveyor_Control_System* (controls, fault lamps, timer) and *Conveyor_Status* (animated conveyor image, running/stopped status, moving box animation, timer).

## Architecture

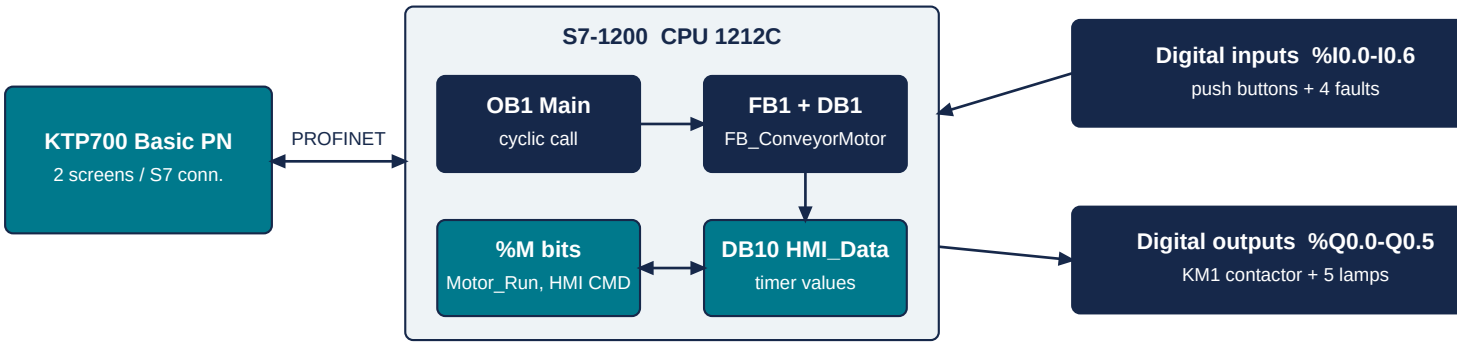



Figure 1 - Controller, HMI and field I/O.

## Operating summary

| Event | System response |
|---|---|
| START (panel or HMI) | Motor_Run seals in, %Q0.0 energises KM1, 25 s timer starts, HMI shows RUNNING. |
| STOP (panel or HMI) | Motor_Run drops, timer resets. HMI STOP command is a momentary bit (released on button release). |
| EM STOP (HMI) | Sets HMI_Stop_CMD and clears HMI_Start_CMD. Latches until START or RESET clears it. |
| Any fault input | Individual lamp on, Main_Fault latched, Main Fault lamp on, motor stopped, HMI start/stop commands cleared. |
| RESET (panel or HMI) | Main_Fault reset (only effective once the fault inputs are clear); system ready to start again. |
| Timer complete | Motor_Run reset automatically; Timer_Complete cleared once the motor is off. |

# 2. Hardware and wiring

The wiring diagram below shows the intended field wiring for the S7-1200 DC/DC/DC controller: 24 V DC supply, push buttons and fault contacts on the digital inputs, KM1 contactor coil and fault lamps on the digital outputs, and the three-phase motor power circuit (MCCB, KM1, overload relay). See section 7 for how the diagram relates to the addressing used in the PLC program.

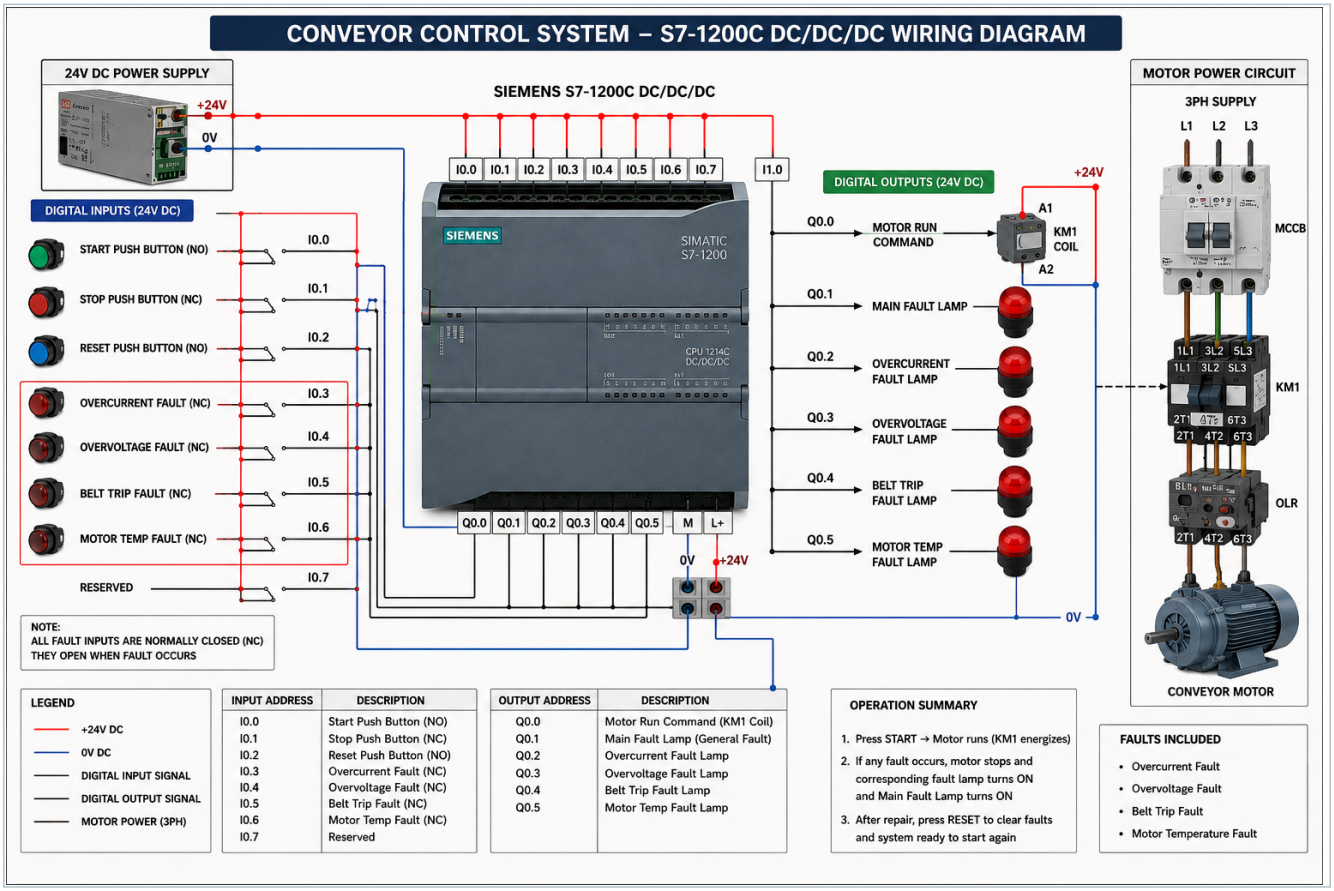



Figure 2 - Conveyor control system wiring diagram (illustrative).

# 3. PLC program

## 3.1 Program blocks

| Block | Type | Language | Purpose |
|---|---|---|---|
| Main [OB1] | Organisation block | LAD | Calls FB_ConveyorMotor with the I/O tags; drives %Q0.0 from %M0.0 (Motor_Run). |
| FB_ConveyorMotor [FB1] | Function block | LAD | All control logic: start/stop, timer, fault lamps, fault latch, shutdown, auto-stop, time conversion. |
| FB_ConveyorMotor_DB [DB1] | Instance DB | - | Instance data of FB1 (static variables Timer_Complete, Main_Fault, Timer_Value, Timer_MS). |
| Run_Timer [DB2] | IEC timer DB | - | TON instance used in Network 2. |
| HMI_Data [DB10] | Global DB | - | Timer value and seconds value for the HMI. Optimised block access is **disabled** so the HMI can use absolute addressing. |

## 3.2 PLC tag table (Conveyor_Tags)

| Tag name | Address | Type | Description |
|---|---|---|---|
| `Start_PB` | `%I0.0` | Bool | Start push button |
| `Stop_PB` | `%I0.1` | Bool | Stop push button (logic uses an NC contact: motor can run while input is FALSE) |
| `Overcurrent_Fault` | `%I0.2` | Bool | Overcurrent fault input (TRUE = fault) |
| `Overvoltage_Fault` | `%I0.3` | Bool | Overvoltage fault input (TRUE = fault) |
| `Belt_Trip_Fault` | `%I0.4` | Bool | Belt trip fault input (TRUE = fault) |
| `Motor_Temp_Fault` | `%I0.5` | Bool | Motor temperature fault input (TRUE = fault) |
| `Reset_PB` | `%I0.6` | Bool | Fault reset push button |
| `Main_Conveyor_Motor` | `%Q0.0` | Bool | Motor run command (KM1 contactor coil) |
| `Overcurrent_Fault_Lamp` | `%Q0.1` | Bool | Overcurrent lamp |
| `Overvoltage_Fault_Lamp` | `%Q0.2` | Bool | Overvoltage lamp |
| `Belt_Trip_Fault_Lamp` | `%Q0.3` | Bool | Belt trip lamp |
| `Motor_Temp_Fault_Lamp` | `%Q0.4` | Bool | Motor temperature lamp |
| `Main_Fault_Lamp` | `%Q0.5` | Bool | Main fault lamp |
| `Motor_Run` | `%M0.0` | Bool | Motor running (FB1 output, copied to %Q0.0 in OB1) |
| `Timer_Complete` | `%M0.1` | Bool | Timer complete flag (watch-table tag; FB1 uses its own static #Timer_Complete) |
| `Main_Fault` | `%M0.2` | Bool | Main fault latch (tag-table entry; FB1 uses its own static #Main_Fault) |
| `HMI_Start_CMD` | `%M0.3` | Bool | HMI start command (latched until STOP / EM STOP / RESET / fault) |
| `HMI_Stop_CMD` | `%M0.4` | Bool | HMI stop / emergency-stop command |
| `HMI_Reset_CMD` | `%M0.5` | Bool | HMI reset command (momentary) |

## 3.3 FB_ConveyorMotor [FB1] interface

| Section | Name | Data type | Default |
|---|---|---|---|
| Input | `Start_PB, Stop_PB, Reset_PB` | Bool | false |
| Input | `HMI_Start_CMD, HMI_Stop_CMD, HMI_Reset_CMD` | Bool | false |
| Input | `Overcurrent_Fault, Overvoltage_Fault, Belt_Trip_Fault, Motor_Temp_Fault` | Bool | false |
| Output | `Overcurrent_Fault_Lamp, Overvoltage_Fault_Lamp, Belt_Trip_Lamp, Motor_Temp_Lamp, Main_Fault_Lamp` | Bool | false |
| Output | `Motor_Run` | Bool | false |
| Static | `Timer_Complete, Main_Fault` | Bool | false |

| Section | Name | Data type | Default |
|---|---|---|---|
| Static | Timer_Value | Time | T#0ms |
| Static | Timer_MS | DInt | 0 |

All interface variables are non-retain and accessible from HMI / OPC UA / Web API.

## 3.4 HMI_Data [DB10]

| Name | Type | Address | Description |
|---|---|---|---|
| Timer_Value | Time | %DB10.DBD0 | Elapsed time (ET) of the Run_Timer TON, written directly by Network 2 |
| Temp_Timer_DInt | DInt | %DB10.DBD4 | Elapsed time in whole seconds, result of the DIV in Network 9; shown on both HMI screens |

## 3.5 OB1 - Main

OB1 contains a single network: the call of **FB_ConveyorMotor** with instance DB1, all physical inputs and HMI command bits wired to the matching FB pins, the five lamp outputs written straight to %Q0.1-%Q0.5, and Motor_Run routed to %M0.0. Below the call, a contact on %M0.0 drives the motor output coil %Q0.0.

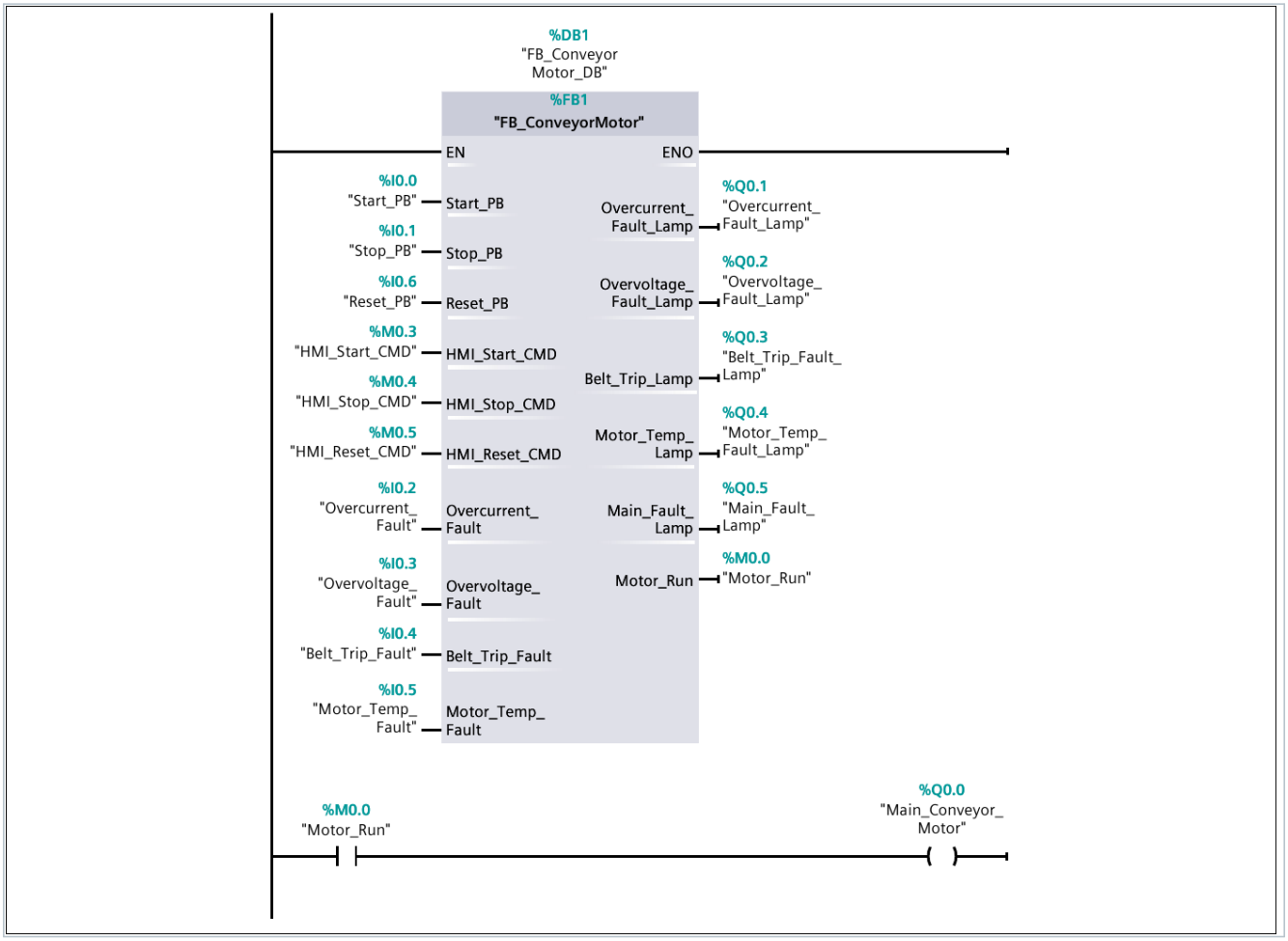
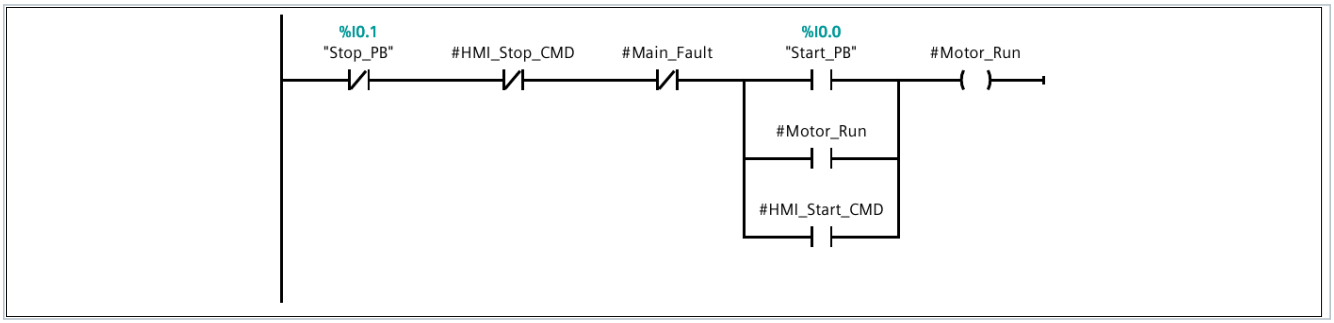



Figure 3 - Main [OB1], Network 1: FB call and motor output.

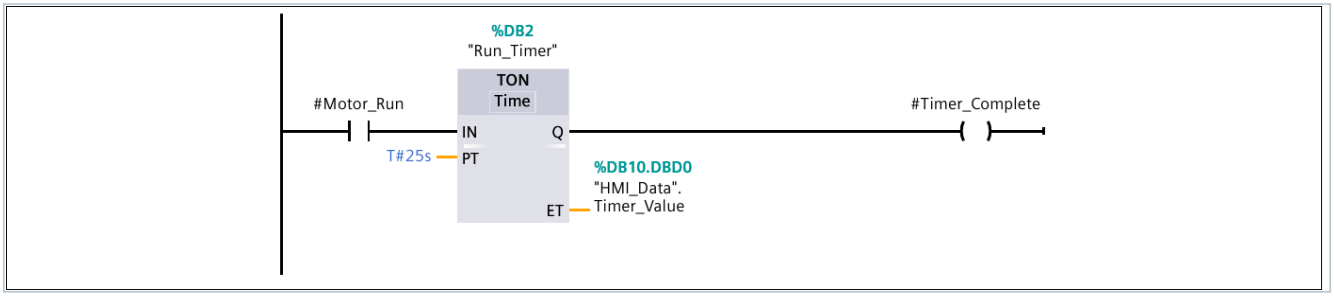
## 3.6 FB_ConveyorMotor [FB1] - ladder networks

### Network 1: Motor Start / Stop Control



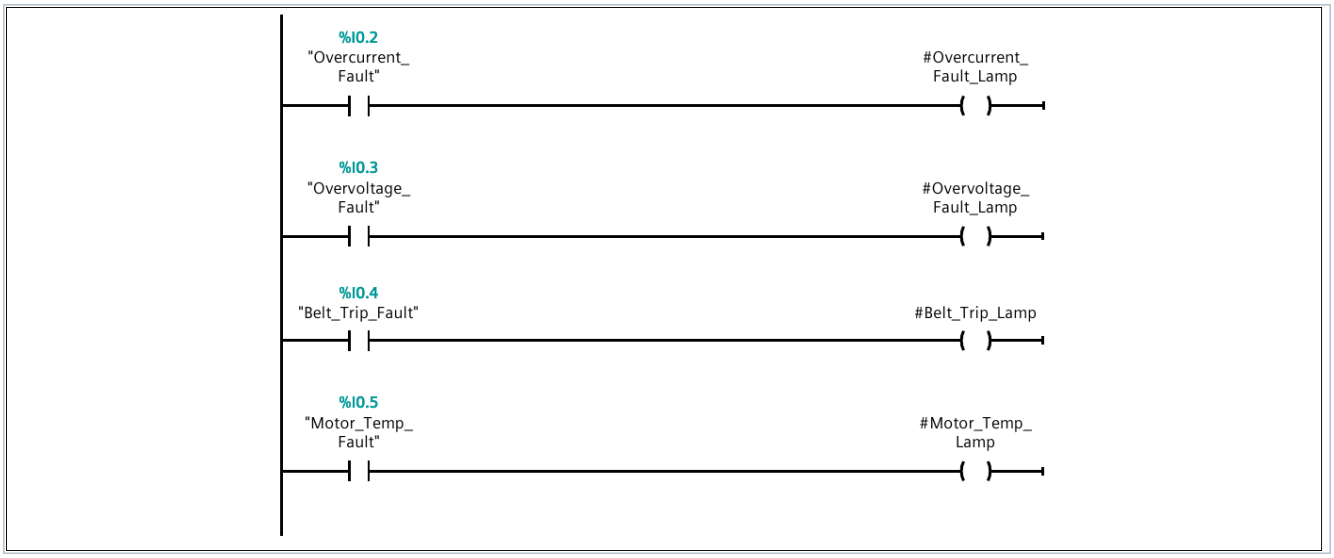

Series path: Stop_PB (NC contact), #HMI_Stop_CMD (NC), #Main_Fault (NC), then a parallel group of Start_PB, #Motor_Run (seal-in) and #HMI_Start_CMD into the #Motor_Run coil. A fault or either stop command breaks the path.

### Network 2: Motor Run Timer



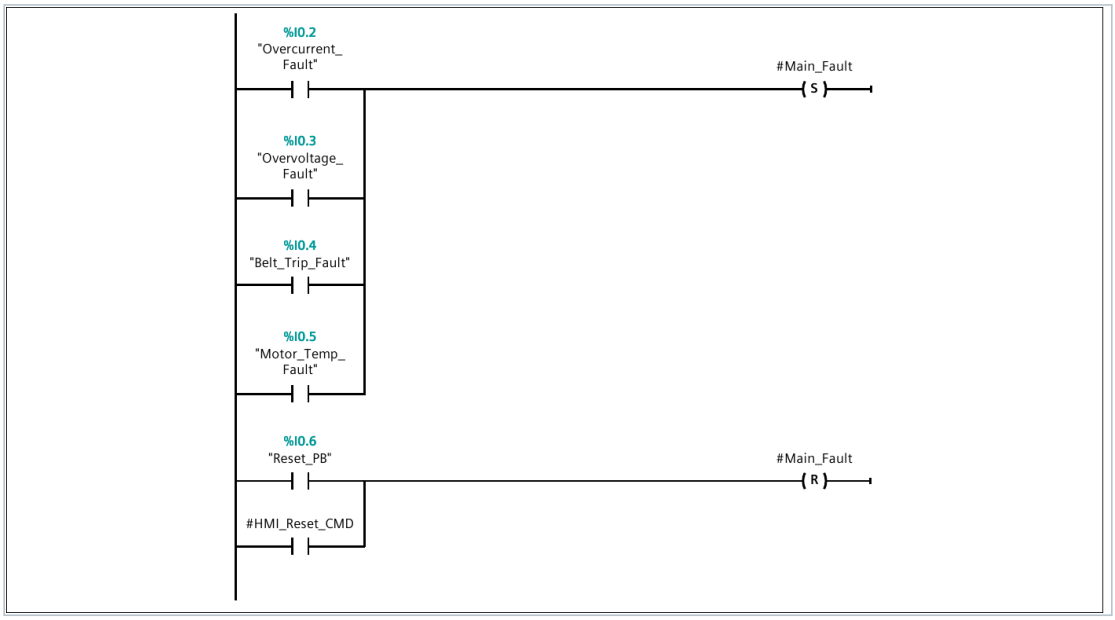

TON "Run_Timer" (DB2) is enabled by #Motor_Run with PT = T#25s. Q drives #Timer_Complete and the elapsed time (ET) is written straight to "HMI_Data".Timer_Value (%DB10.DBD0).

### Network 3: Individual Fault Lamps



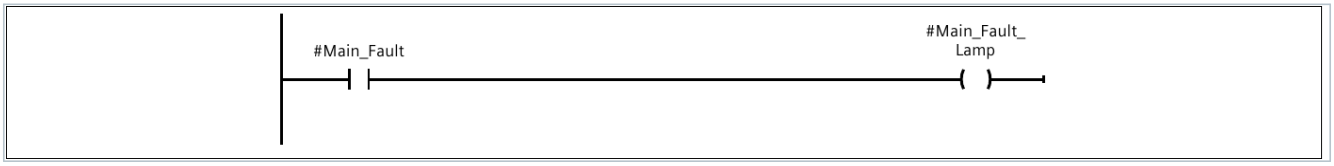

Each fault input (%I0.2-%I0.5) drives its own lamp output bit, so every active fault is indicated independently.

### Network 4: Main Fault Latch SET / RESET




Any of the four fault inputs (parallel branches) SETs #Main_Fault. Reset_PB (%I0.6) or #HMI_Reset_CMD RESETs it. The reset is placed after the set so it takes priority while the reset input is active.

### Network 5: Main Fault Indication



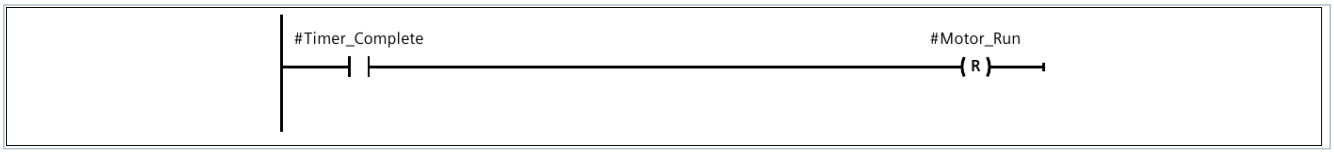

#Main_Fault drives #Main_Fault_Lamp (%Q0.5).

### Network 6: Fault Motor Shutdown



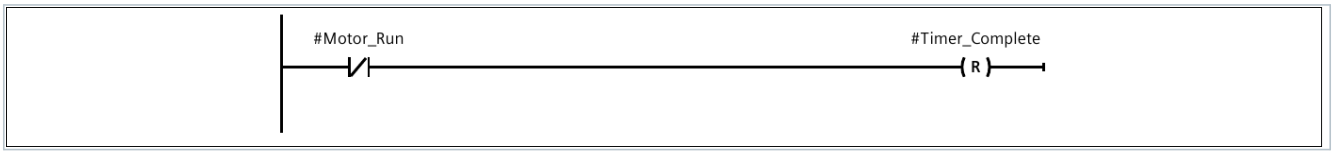

#Main_Fault resets #Motor_Run and also resets the HMI command bits %M0.3 (HMI_Start_CMD) and %M0.4 (HMI_Stop_CMD), so the motor cannot restart by itself after the fault is reset.

### Network 7: Auto Stop on Timer Complete



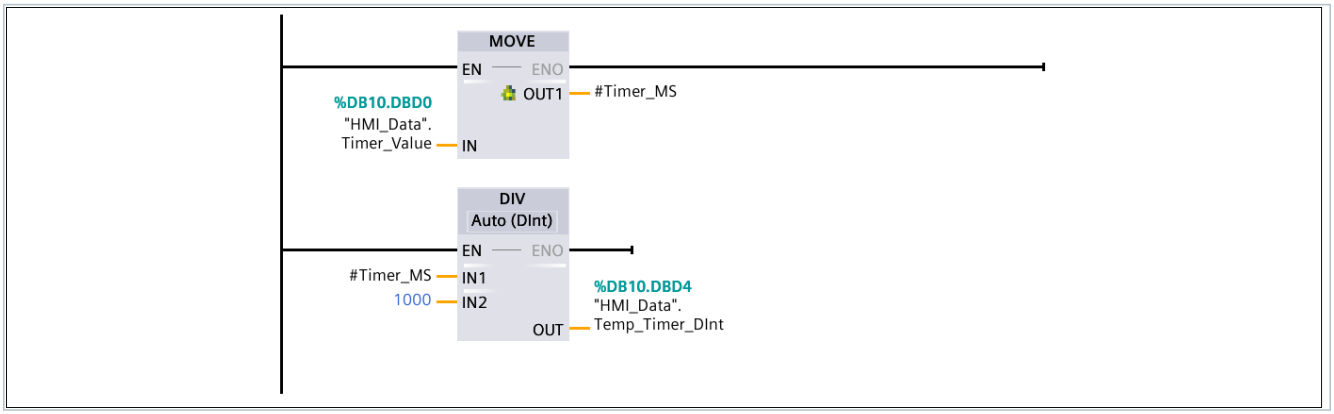

#Timer_Complete resets #Motor_Run when the 25 s run time has elapsed.

### Network 8: Timer Complete Reset




With the motor off (NC contact on #Motor_Run), #Timer_Complete is reset so the next run starts clean.

### Network 9: Timer Seconds Conversion




MOVE copies "HMI_Data".Timer_Value (Time) into #Timer_MS (DInt) using implicit conversion. DIV (DInt) divides #Timer_MS by 1000 and writes whole seconds to %DB10.DBD4 for the HMI timer field.

# 4. PLC simulation evidence

The program was tested online against S7-PLCSIM V21. The captures below show a fault simulation in the PLCSIM SimTable, the OB1 and FB1 networks monitored online (green = energised), and the TIA Portal watch table with live values.

## 4.1 Fault simulation (S7-PLCSIM SimTable)

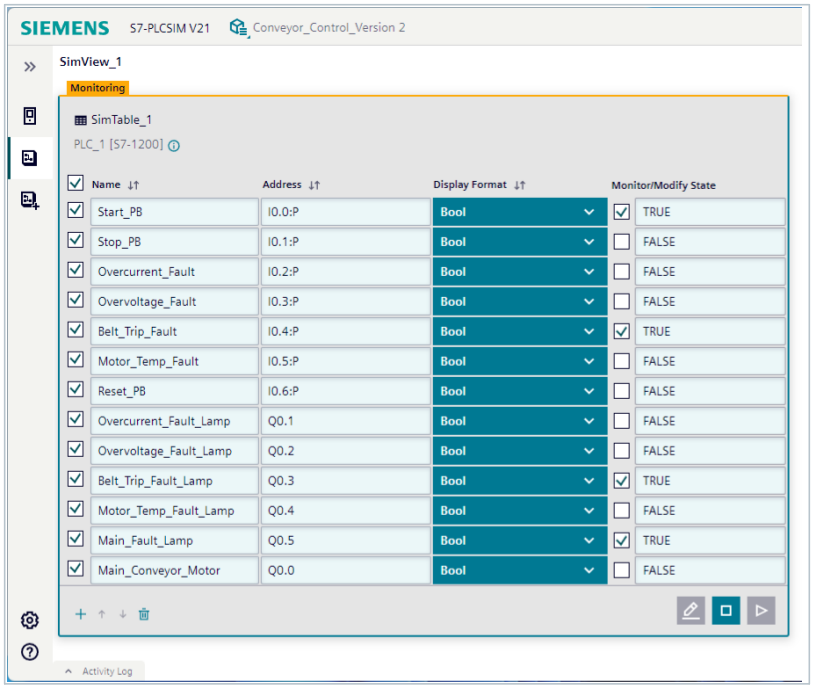



Figure 4 - Start_PB and Belt_Trip_Fault set TRUE: Belt_Trip_Fault_Lamp (Q0.3) and Main_Fault_Lamp (Q0.5) are on and Main_Conveyor_Motor (Q0.0) is FALSE.

## 4.2 Main [OB1] online

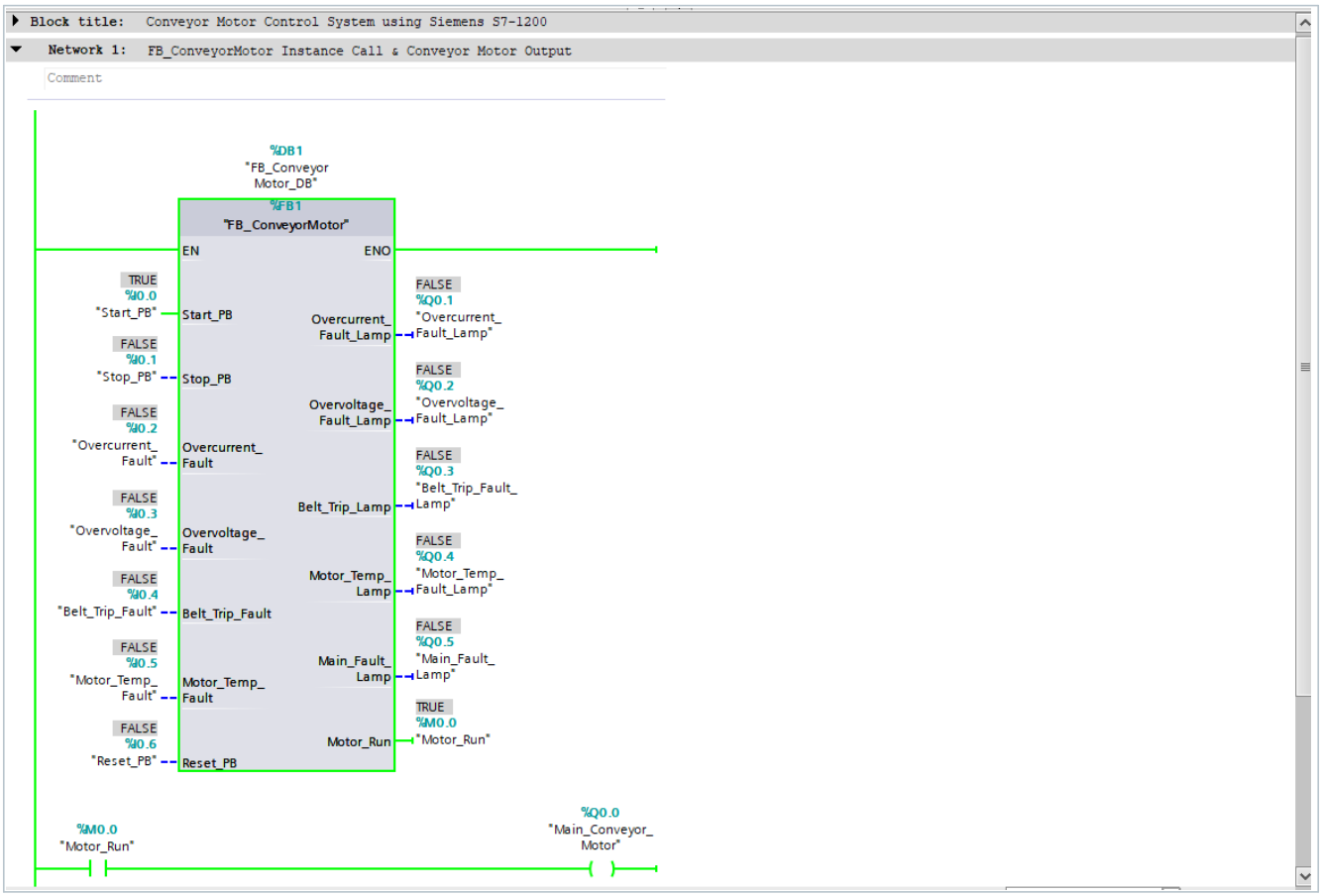



Figure 5 - OB1 monitored online: inputs, lamp outputs and Motor_Run (%M0.0) values on the FB call.

## 4.3 FB_ConveyorMotor [FB1] online

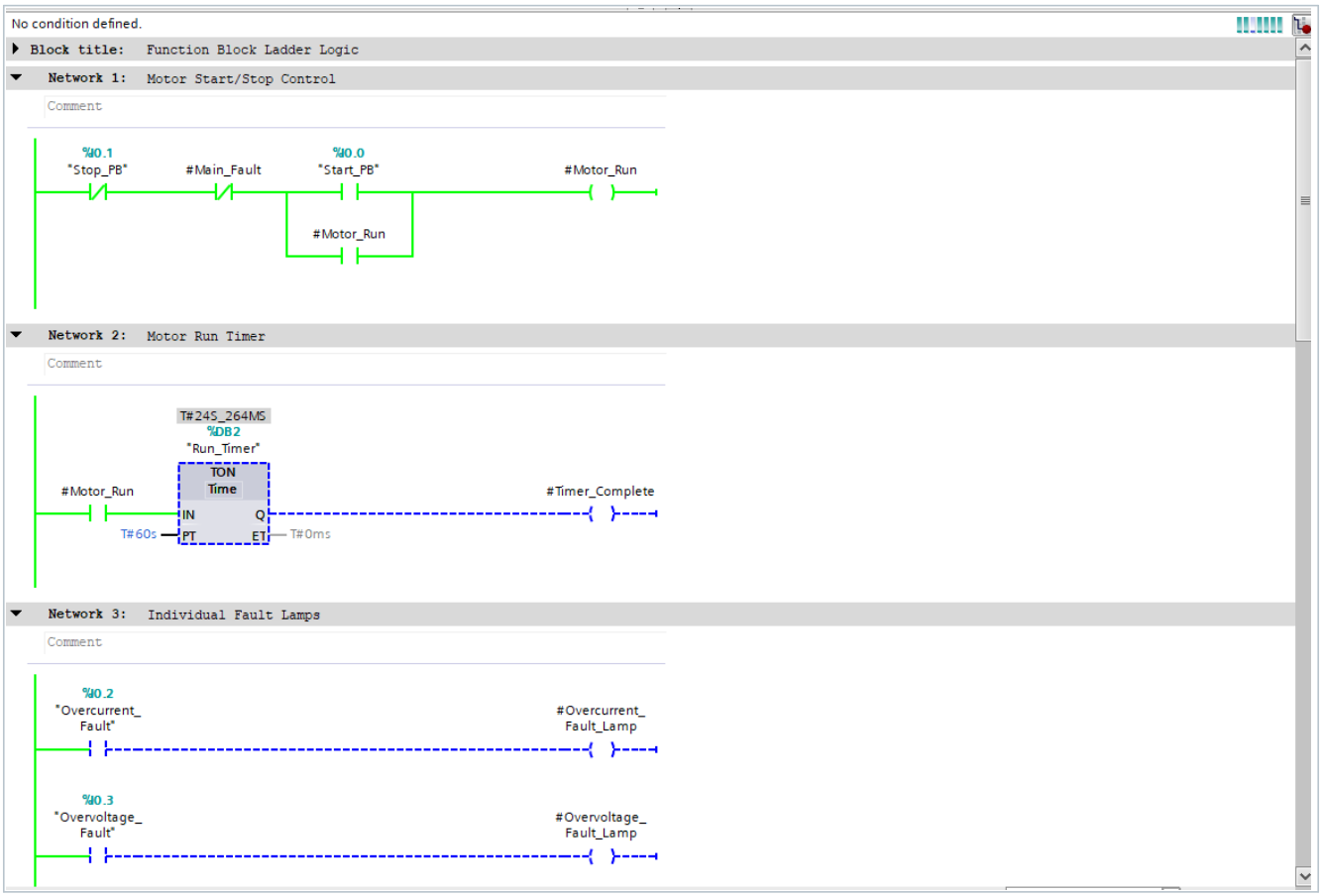



Figure 6 - FB1 networks 1-3 monitored online. Motor_Run is sealed in and the run timer is counting. This capture was taken with an earlier timer preset (T#60s); the exported program uses T#25s.

## 4.4 Watch table with live values

| | Name | Address | Display format | Monitor value | Modify value | Comment | Tag comment |
|---|---|---|---|---|---|---|---|
| 1 | "Start_PB" | %I0.0 | Bool | TRUE | TRUE | | |
| 2 | "Stop_PB" | %I0.1 | Bool | FALSE | | | |
| 3 | "Overcurrent_Fault" | %I0.2 | Bool | FALSE | | | |
| 4 | "Overvoltage_Fault" | %I0.3 | Bool | FALSE | | | |
| 5 | "Belt_Trip_Fault" | %I0.4 | Bool | FALSE | | | |
| 6 | "Motor_Temp_Fault" | %I0.5 | Bool | FALSE | | | |
| 7 | "Reset_PB" | %I0.6 | Bool | FALSE | | | |
| 8 | "Motor_Run" | %M0.0 | Bool | TRUE | | | |
| 9 | "Timer_Complete" | %M0.1 | Bool | FALSE | | | |
| 10 | | %M0.3 | Bool | FALSE | | | |
| 11 | "Main_Conveyor_Motor" | %Q0.0 | Bool | TRUE | | | |
| 12 | "Overcurrent_Fault_Lamp" | %Q0.1 | Bool | FALSE | | | |
| 13 | "Overvoltage_Fault_Lamp" | %Q0.2 | Bool | FALSE | | | |
| 14 | "Belt_Trip_Fault_Lamp" | %Q0.3 | Bool | FALSE | | | |
| 15 | "Motor_Temp_Fault_Lamp" | %Q0.4 | Bool | FALSE | | | |
| 16 | "Main_Fault_Lamp" | %Q0.5 | Bool | FALSE | | | |
| 17 | | <Add new> | | | | | |

Figure 7 - Watch table "Watch_Conveyor": Start_PB TRUE, Motor_Run (%M0.0) TRUE and Main_Conveyor_Motor (%Q0.0) TRUE, all faults FALSE.

# 5. HMI

## 5.1 Panel and connection

| Setting | Value |
|---|---|
| Panel | KTP700 Basic PN (800 x 480 px) |
| Connection | HMI_Connection_1, SIMATIC S7 1200 |
| HMI IP address | 192.168.0.2 |
| PLC IP address | 192.168.0.1 |
| Screens | 1 - Conveyor_Control_System \| 2 - Conveyor_Status |
| Text list | Motor_Status_Text (0 = STOPPED, 1 = RUNNING) |

## 5.2 HMI tag table

| HMI tag | Connected to | Address | Used for |
|---|---|---|---|
| `HMI_Motor_Run` | `Motor_Run` | `%M0.0` | Motor graphics, status lamp, status text (both screens). 100 ms. |
| `HMI_Start_CMD` | `HMI_Start_CMD` | `%M0.3` | START button set bit. 100 ms. |
| `HMI_Stop_CMD` | `HMI_Stop_CMD` | `%M0.4` | STOP / EM STOP set bit. 100 ms. |
| `HMI_Reset_CMD` | `HMI_Reset_CMD` | `%M0.5` | RESET button set bit. 100 ms. |
| `HMI_Overcurrent` | `Overcurrent_Fault` | `%I0.2` | Overcurrent lamp. 1 s. |
| `HMI_Overvoltage` | `Overvoltage_Fault` | `%I0.3` | Overvoltage lamp. 1 s. |
| `HMI_Belt_Trip` | `Belt_Trip_Fault` | `%I0.4` | Belt trip lamp. 1 s. |
| `HMI_Motor_Temp` | `Motor_Temp_Fault` | `%I0.5` | Motor temperature lamp. 1 s. |
| `HMI_Fault_Latch` | `Main_Fault` | `%M0.2` | Main Fault lamp. 1 s. |
| `HMI_Timer_Sec` | `HMI_Data (DB10)` | `symbolic` | Timer I/O field, seconds (symbolic access so DB changes do not break the tag). 100 ms. |
| `Main_Conveyor_Motor` | `Main_Conveyor_Motor` | `%Q0.0` | Box animation visibility on screen 2 (used directly). |
| `HMI_Box_Position` | `(internal)` | `-` | Box position 0-100, driven by SimulateTag on screen 2. |

Physical inputs cannot be written from the HMI, so the buttons use the dedicated memory bits %M0.3-%M0.5. Button actions use SetBit / ResetBit instead of SetBitWhileKeyPressed, which proved unreliable in PLCSIM.

## 5.3 Screen 1 - Conveyor_Control_System

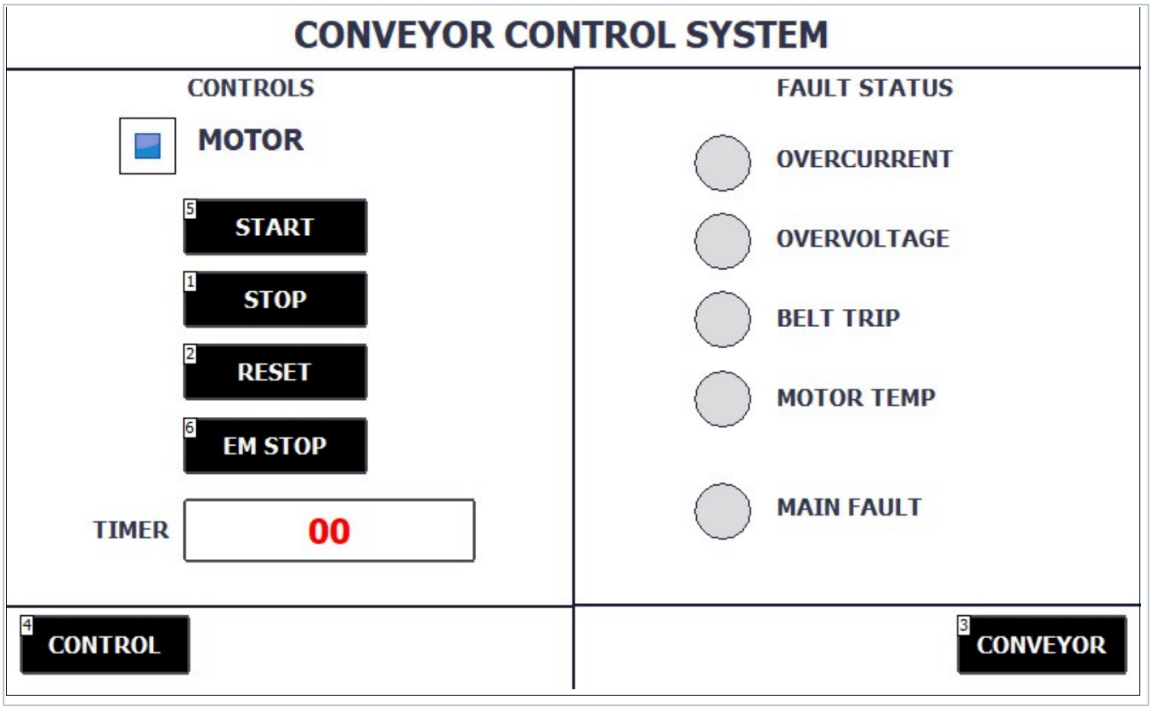



Figure 8 - Screen 1 hardcopy from TIA Portal (design view).

| Element | Behaviour |
|---|---|
| MOTOR indicator | Two stacked graphics. Green graphic visible when HMI_Motor_Run = 1, red graphic visible when HMI_Motor_Run = 0. |
| START button | Press: SetBit HMI_Start_CMD, ResetBit HMI_Stop_CMD. No release event, so the command latches until cleared. |
| STOP button | Press: ResetBit HMI_Start_CMD, SetBit HMI_Stop_CMD. Release: ResetBit HMI_Stop_CMD. |
| RESET button | Press: SetBit HMI_Reset_CMD, ResetBit HMI_Stop_CMD, ResetBit HMI_Start_CMD. Release: ResetBit HMI_Reset_CMD. |
| EM STOP button | Press: SetBit HMI_Stop_CMD, ResetBit HMI_Start_CMD. No release event, so it latches until START or RESET. |
| TIMER I/O field | Output field, HMI_Timer_Sec, decimal, 2 digits (pattern 99), red 24 px bold. |
| Fault lamps (circles) | Overcurrent, Overvoltage, Belt Trip, Motor Temp: grey (value 0) / red (value 1). Main Fault: grey / orange (value 1). |
| CONVEYOR button | Click: ActivateScreen Conveyor_Status. |

## 5.4 Screen 2 - Conveyor_Status

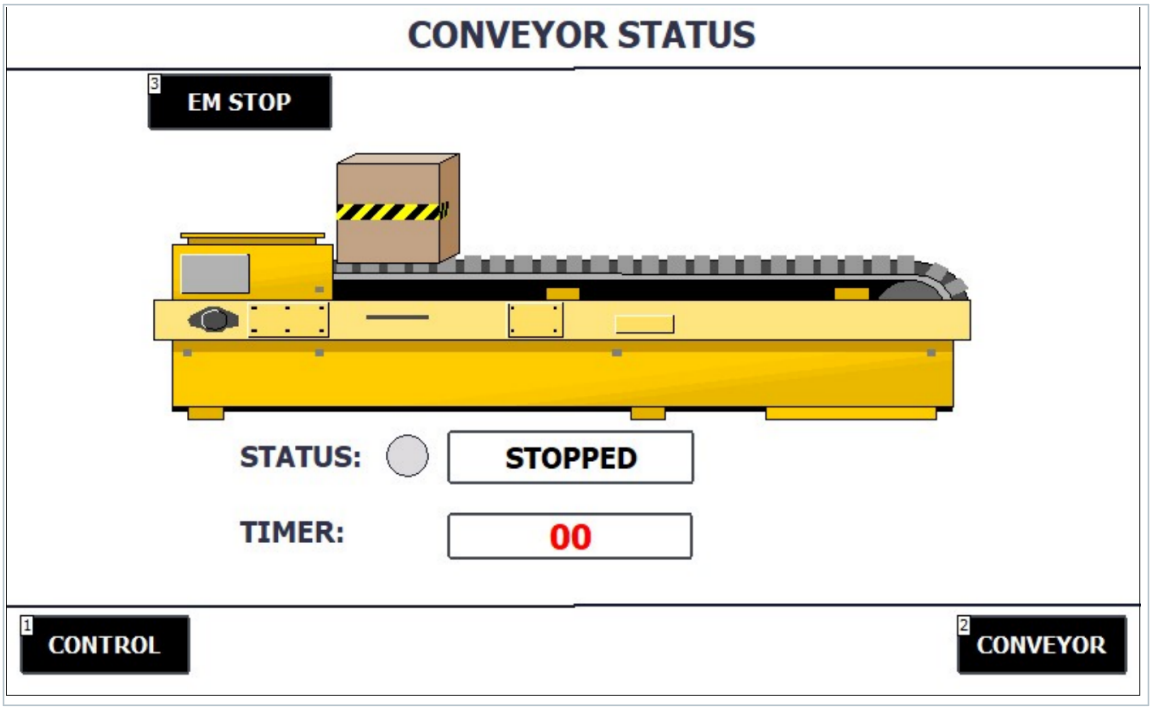



Figure 9 - Screen 2 hardcopy from TIA Portal (design view).

| Element | Behaviour |
|---|---|
| Conveyor image | Appearance by HMI_Motor_Run: dark grey (82, 81, 82) when 0, yellow (255, 215, 0) when 1. |
| Box graphic (moving) | Visible when Main_Conveyor_Motor = 1. Horizontal movement follows HMI_Box_Position (0-100) over 346 px. |
| Box graphic (parked) | Visible when Main_Conveyor_Motor = 0, shown at the start of the belt. |
| Screen event "Loaded" | SimulateTag HMI_Box_Position: cycle 1 s, minimum 0, maximum 100, step 1. |
| STATUS lamp | Circle: red when HMI_Motor_Run = 0, green when 1. |
| STATUS text | Symbolic I/O field on HMI_Motor_Run, text list Motor_Status_Text: STOPPED / RUNNING. |
| TIMER I/O field | Output field, HMI_Timer_Sec, decimal, pattern 99. |
| EM STOP button | Press: SetBit HMI_Stop_CMD, ResetBit HMI_Start_CMD (same as screen 1). |
| CONTROL button | Click: ActivateScreen Conveyor_Control_System. |

## 5.5 Runtime simulation captures

Captured from the recorded WinCC runtime simulation (Conveyor_Control_HMI_Simulation.mp4, included in the repository).

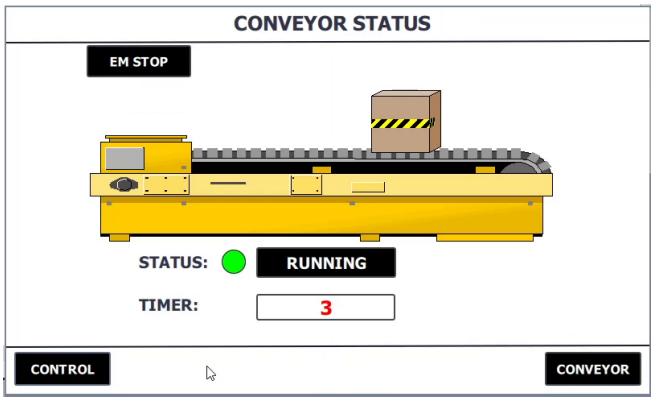



Figure 10 - Status screen, motor RUNNING, box in motion, timer counting.

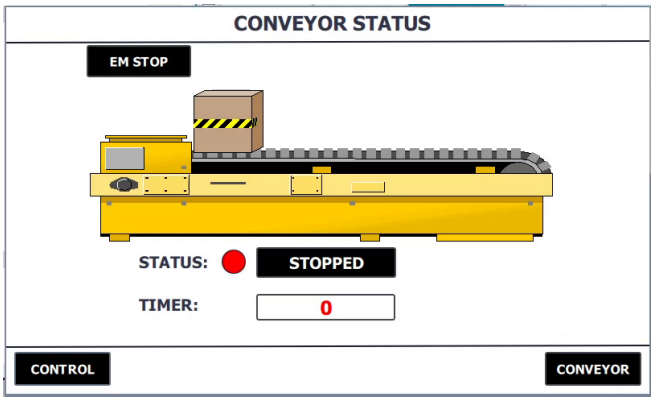



Figure 11 - Status screen after a fault: STOPPED, red status lamp, timer 0.

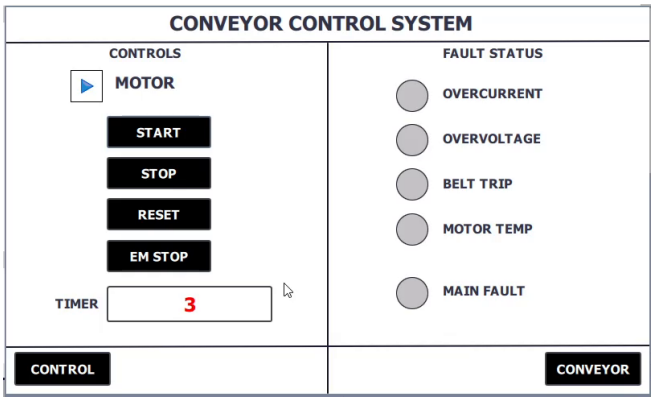



Figure 12 - Control screen with motor running (blue play symbol), no faults.

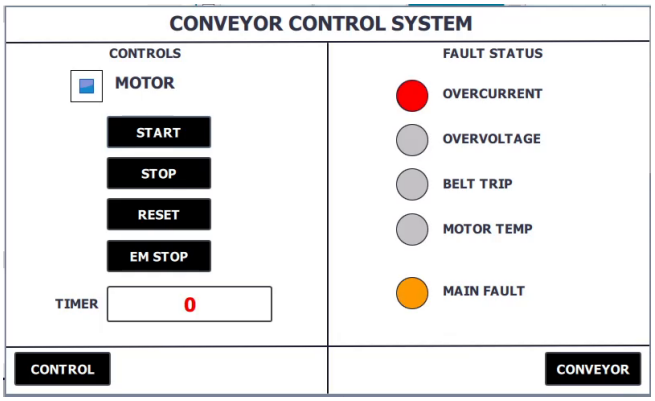



Figure 13 - Overcurrent fault: Overcurrent and Main Fault lamps active, timer 0.

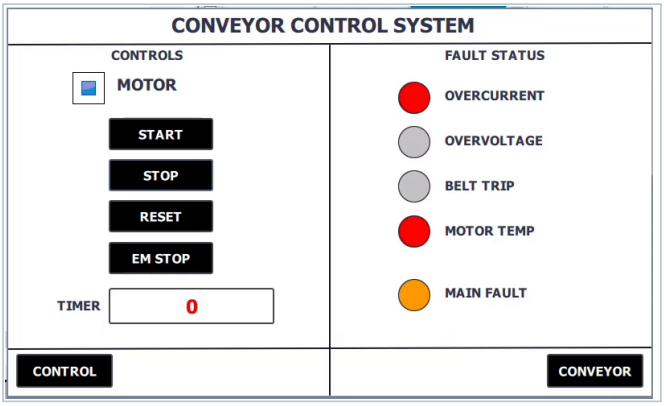



Figure 14 - Multiple faults: Overcurrent, Motor Temp and Main Fault lamps active.

# 6. Operating sequences

| Sequence | Steps |
|---|---|
| Normal start / stop | 1. HMI START (or %I0.0) sets the start path in Network 1. 2. Motor_Run seals in, %M0.0 and %Q0.0 go TRUE. 3. Timer counts from 0 to 25 s; HMI shows seconds. 4. STOP (panel or HMI) opens the path; Motor_Run and the timer reset. |
| Timer auto-stop | 1. Timer reaches T#25s, #Timer_Complete TRUE. 2. Network 7 resets #Motor_Run. 3. Network 8 resets #Timer_Complete once the motor is off. |
| Emergency stop | 1. EM STOP sets HMI_Stop_CMD and clears HMI_Start_CMD. 2. Network 1 is broken by the NC contact on #HMI_Stop_CMD. 3. START or RESET clears HMI_Stop_CMD before a new start. |
| Fault | 1. A fault input goes TRUE. 2. Its lamp lights (Network 3), #Main_Fault is SET (Network 4) and the Main Fault lamp lights (Network 5). 3. Network 6 resets Motor_Run, HMI_Start_CMD and HMI_Stop_CMD; the motor stops and cannot restart while the latch is set. |
| Reset | 1. Clear the fault condition. 2. Press RESET (panel or HMI). 3. #Main_Fault resets, lamps clear. 4. Press START to run again. |

# 7. Design notes

## 7.1 Design decisions

- **#variables inside the FB.** Logic inside FB1 uses interface/static variables. Mixing global %M addresses with FB output pins caused memory conflicts during development.
- **Only Motor_Run is routed through OB1.** It leaves the FB on a pin to %M0.0 and then drives %Q0.0; the fault lamps are written directly to %Q0.1-%Q0.5.
- **DB10 optimised block access is disabled** so the HMI can reach the data by absolute address; the timer tag is accessed symbolically so DB edits do not shift it.
- **Dedicated %M command bits for the HMI** because physical %I inputs cannot be written from the panel.
- **SetBit / ResetBit instead of SetBitWhileKeyPressed** for reliability in PLCSIM.
- **Network 9 uses MOVE + DIV** because the TIME_TO_DINT instruction was not available in this TIA Portal environment.
- **KTP700 Basic limits:** images cannot be animated natively (the moving box is a separate graphic driven by SimulateTag); the Siemens branding and F1-F8 keys are fixed hardware elements.

## 7.2 Points to be aware of

- **Wiring diagram vs program addressing.** The wiring diagram is illustrative: its CPU label, input/output order (for example Reset on I0.2, Main Fault lamp on Q0.1) and NC drawing of the stop and fault contacts differ from the addresses and polarity in the program. The program treats the four fault inputs as active-TRUE and uses an NC contact for Stop_PB. Before wiring real hardware, make the diagram, tag table and contact polarity consistent.
- **Start command after auto-stop.** HMI_Start_CMD is cleared by STOP, EM STOP, RESET and faults, but not by the timer auto-stop in Network 7. If the motor was started from the HMI, check how the program behaves after the timer completes and consider clearing HMI_Start_CMD there.
- **Timer preset.** The exported program and the recorded simulation use T#25s; the online FB1 capture in section 4 shows an earlier T#60s version.